Exercise 10: Convolutional Neural Networks

Task 1:

a) Explain why it is not necessary to manually define which specific filters (e.g. edge detection filters, sharpening filters) are used in the individual convolutional layers when training a CNN.
b) Explain why CNN architectures usually consist of *multiple* convolutional layers.
c) Why is CNN (in theory) better suited for image processing compared to an MLP?
d) If one convolution layer applies seven 3x3x3 filters to the input, what is then the value for d in the size of the filters in the subsequent convolution layer 5x5xd?

Task 2:

a) Convolution: Calculate the output feature matrix S of a convolution layer given the following Input I, filter (aka kernel) K, stride s, and padding P.

Example for demonstration:

$$I=\begin{bmatrix}1&2&3&0\\0&1&2&3\\3&0&1&2\\2&3&0&1\end{bmatrix},\ K=\begin{bmatrix}0&1&0\\1&0&1\\0&1&0\end{bmatrix},\ s=3,P=1\rightarrow\ S=\begin{bmatrix}2&6\\6&2\end{bmatrix}$$

Your task:

$$I=\begin{bmatrix}2&1&0&3&1\\1&2&1&0&2\\3&1&2&1&0\\0&2&1&3&1\\1&0&2&1&2\end{bmatrix},\ K=\begin{bmatrix}0&1&0\\1&-4&1\\0&1&0\end{bmatrix},\ s=2,P=1$$

b) Max Pooling: Calculate the output feature matrix S for a max pooling layer given the following Input I, stride s, Kernel size F, and padding P.

Example for demonstration:

$$I=\begin{bmatrix}1&2&3&0\\0&1&2&3\\3&0&1&2\\2&3&0&1\end{bmatrix},\ s=2,P=0,\ F=2\rightarrow\ S=\begin{bmatrix}2&3\\3&2\end{bmatrix}$$

Your task:

$$I=\begin{bmatrix}1&3&2&4\\5&6&1&2\\9&8&3&1\\4&7&2&0\end{bmatrix},\ s=2,P=0,\ F=2$$

Task 3:

Revisit the **MNIST digits** dataset, but this time train a CNN instead of an MLP for the classification. Compare the performance and number of parameters to your models from the previous weeks.